sanas South African National Accreditation System	**TECHNICAL REQUIREMENTS OF ISO/IEC 17025:2017**	Organisation's SANAS No/s.

Date/s of evaluation	
Assessor/s & Observers	
Laboratory	
Area / field of operation	
Laboratory Representative	

This report covers the following:

Document Review only		Implementation on Site Visit only		Document Review and Site Visit		Other	

REQUIREMENTS & COMMENTS. ***Compliance = C, Non-compliance = NC***

NB1: *References to ISO/IEC 17025:2017 are in italics & indicated as Std. The order of assessment need not follow the order of the checklist. Assessors are expected to know & have the standard, this worksheet is designed as guidance to prompt detailed recording of the process.*

REFER TO ISO/IEC 17025:2017 FOR DETAIL AND FOR CLARIFICATION NOTES.

NOTE 1: *For **CAB's comments**: The CAB must provide information on how requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*

NOTE 2: *For **Assessor's Comments**: The Assessor must provide information on the CAB's conformity with the requirements*

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CLAUSE	ISO/IEC 17025:2017 REQUIREMENTS How are the following addressed / implemented	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
6	**RESOURCE REQUIREMENTS**			
6.1	Availability of personnel, facilities, equipment, systems and support services necessary for the management and performance laboratory activities.	• We maintain a resource-planning register. For each test group (e.g. proximate analysis, ultimate analysis, ash, moisture), we assign sufficient qualified staff, ensure dedicated lab space with required environmental control (HVAC, extraction), maintain an asset inventory (equipment register, software, consumables). We monitor workload and staffing levels quarterly and adjust. **How do we monitor workload and staffing levels quarterly and adjust accordingly**		
6.2	**Personnel:** *How are the following addressed/implemented?*			
6.2.1	Personnel, either internal or external, that could influence the laboratory activities act impartially, are competent and work in accordance with the laboratory's management system.	• We have a conflict-of-interest declaration ASMF 4.2.1 . All technical staff, whether permanent or contract, sign a confidentiality declaration upon boarding. External subcontractors must meet competence criteria as outlined in the documents 6.2 Selection, evaluation and monitoring of subcontractors procedure , checklist and register, and their work is covered by our subcontracting agreement under our management system		

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6.2.2	Competence requirements are documented for each function influencing the results of laboratory activities, including requirements for education, qualification, training, technical knowledge, skills and experience.	• Our job descriptions and competency profiles list for each role the minimum qualification required, experience, required training. • Personnel competency is comprised of initial training, technical assessment(theory e.g CV assessment), witnessing(SANAS F 15-11) and competency authorization.		
6.2.3	Personnel have the competence to perform laboratory activities for which they are responsible and to evaluate the significance of deviations.	• Personnel are trained to detect out-of-control QC flags (e.g. drift, repeat failures, control chart evaluation) and follow deviation protocols. Training includes problem-solving for instrument drift, out-of-control-limit interpretations investigations.		
6.2.4	Laboratory management communicate to personnel their duties, responsibilities and authorities.	• During onboarding, each staff member receives a formal assignment letter stating their responsibilities, and these are reviewed in annual performance appraisal. • It is as per standard practice that no personnel operates a work station without being deemed competent		
6.2.5	The laboratory has procedure(s) and retain records for:	• **"Declaration of Personnel Competence Procedure ASMF 6.2"** addresses points a-f. The lab keeps records: Witnessing evidence, competency assessment records, declaration of authorization (for each method), and annual competence refresher witnessing will serve for monitoring of personnel competence.		

CLAUSE	ISO/IEC 17025:2017 REQUIREMENTS *How are the following addressed / implemented*	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
	a) determining the competence requirements;			
	b) selection of personnel;			
	c) training of personnel;			
	d) supervision of personnel;			
	e) authorization of personnel;			
	f) monitoring of competence of personnel.			
6.2.6	Personnel authorized to perform specific laboratory activities, including but not limited to, the following:	As per job descriptions, lab managers/supervisors onboarding the organization are tasked to develop, modify (where necessary) and validate methods employed in the laboratory, reviewing and authorization of results. Only personnel who satisfy competence and authorization criteria are allowed to perform or authorize work as outlined in each SOP. This also satisfies points A through C		
	a) development, modification, verification and validation of methods;			
	b) analysis of results, including statements of conformity or opinions and interpretations;			
	c) report, review and authorization of results.			
6.3	**Facilities and environmental conditions:** *How are the following addressed/implemented?*			

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6.3.1 ***Note in Std.***	Facilities and environmental conditions are suitable for the laboratory activities and are not adversely affecting the validity of results.	• The laboratory is located in a controlled building, with separation of the sample preparation area, analysis room, storage and furnace room for easier control and monitoring of the different conditions associated with each area. See attached floor plan.		
6.3.2	Requirements for facilities and environmental conditions necessary for the performance of the laboratory activities are documented.	• In each SOP, the organization has included environmental requirements needed to perform the method of interest together with its tolerance ranges and monitoring frequencies. (e.g. 19-26 °C for the analysis room, 16-20 °C for the storage facility, 25-45 % RH on both), tolerances and monitoring frequency.		
6.3.3	Laboratory monitors, controls and records environmental conditions in accordance with relevant specifications, methods or procedures or where they influence the validity of the results.	• The laboratory has in place aircons in the analysis room and storage facilities that control the temperature to the desired ranges. Hygrometers are present to always read the room temperature and help monitor the relative humidity as well. Deviations outside the acceptable ranges trigger investigation and possible data flagging.		

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6.3.4	Measures to control facilities are implemented, monitored and periodically reviewed and including, but not be limited to: a) access to and use of areas affecting laboratory activities;	• Access to analysis rooms is restricted to authorized personnel. Furnace room is separated from the analysis room for easier control of the analysis room atmosphere and to lessen major temperature fluctuations. Enforcing of cleanliness in analysis room to ensure minimal to no sample contamination (Addresses A through C).		
	b) prevention of contamination, interference or adverse influences on laboratory activities;			
	c) effective separation between areas with incompatible laboratory activities.			
6.3.5	Requirements related to facilities and environmental conditions of this document are met when the laboratory performs laboratory activities at sites or facilities outside its permanent control,	**Not applicable to this CAB since no activities take place outside its permanent facilities.**		
6.4	**Equipment:** *How are the following addressed/implemented?*			
6.4.1 ***Note Std.***	Laboratory has access to equipment required for the correct performance of laboratory activities and which can influence the result.	• The CAB maintains an equipment register which has all critical equipment that have an influence on the quality and validity of results.		

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CLAUSE	ISO/IEC 17025:2017 REQUIREMENTS How are the following addressed / implemented	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
6.4.2	Requirements for equipment of this document are met in those cases where the laboratory uses equipment outside its permanent control.	• Not applicable to this CAB since there are no equipment's the CAB uses outside its permanent control. In the case of subcontracting work, the CAB has a subcontracting procedure ASMF 6.2 that covers critical aspects of the activities on request (e.g equipment calibrations) and thoroughly assesses the capability of the service provider before deciding on engaging with the service provider.		
6.4.3	Procedure for handling, transport, storage, use and planned maintenance of equipment in order to ensure proper functioning and to prevent contamination or deterioration.	**Note:** Ensure all equipment and materials used are verified against factory specifications and logged in the equipment register prior to use. Calibrations (where necessary) and verification of equipment are to be attended prior to use. Calibration services shall only be sourced from ISO/IEC 17025 accredited labs. Verification certificates from manufacturers are accepted for new equipment provided to address traceability. (ISO 17025; Clause 6.4.) This is inserted in method SOP's to raise awareness on personnel for the use of equipment, equipment pretreatment and verification checks to ensure the equipment is fit for purpose. (Addresses 6.4.4 as well)		
6.4.4	Verify that equipment conforms to specified requirements before being placed or returned into service.			

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6.4.5	Equipment used for measurement are capable of achieving the measurement accuracy or measurement uncertainty required to provide a valid result.	• Balances are calibrated by an ISO 17025 accredited facility rendering the balances reliable for use. The CAB has balance calibration certificates as evidence of calibration with its uncertainties included in the certificate. Mass pieces employed within the CAB are calibrated to ensure unerrored verification of balances when they are in use. • See attached examples, **calibration certificate**, **mass pieces**		
6.4.6 ***Note Std.***	Measuring equipment are calibrated when: — the measurement accuracy or measurement uncertainty affects the validity of the reported results, or — calibration of the equipment is required to establish the metrological traceability of the reported result.	• All measurement-critical equipment (balances, temperature monitoring,) are on the calibration schedule. If the calibration status affects the test result, calibration is mandatory.		
6.4.7	Establish a calibration programme, which gets reviewed and adjusted as necessary in order to maintain confidence in the status of calibration.	• Calibrations are done on a yearly schedule by an external ISO 17025 accredited facility. • In the case of Calorific Value instrumentation, calibration is done as and when necessary in line with AS007IW PROCEDURE ON CAL(		
6.4.8	Equipment requiring calibration, or which has a defined period of validity, is labelled, coded, or otherwise identified.	• **All calibrated equipment has calibration stickers, which come from the external supplier.**		

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6.4.9 ***See 7.10***	Equipment subjected to overloading or mishandling, gives questionable results, or shown to be defective or outside specified requirements, are taken out of service.	• Job cards/quotations are required as proof.  38225 AS021PA.pdf		
	Isolated to prevent its use or clearly labelled or marked as being out of service until it has been verified to perform correctly.	• The lab attaches an "out of order" sign for all equipment not in use/decommissioned.		
	Examine the effect of the defect or deviation from specified requirements and shall initiate the management of nonconforming work procedure	• Non-conformance procedure in place to mediate the effects of non-conforming work.  7.17. MANAGEMENT OF NON-CONFORMI		
6.4.10	When intermediate checks are necessary to maintain confidence in the performance of the equipment, these checks are carried out according to a procedure.	• Equipment performance is monitored by the use of control charts upon analysis of certified reference materials. This is outlined in the procedure  ASMF 6.4.10 DEVELOPEMENT AND • Verification of equipment (balances, crushers, pulverisers) before use to assess if the equipment is still fit for purpose.  AS002TF-PULVERISER EFFICIEN		

CLAUSE	ISO/IEC 17025:2017 REQUIREMENTS How are the following addressed / implemented	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
6.4.11	Reference values and correction factors are updated and implemented when calibration and reference material data include reference values or correction factors.	• Correction factors of equipment provided on calibration certificates are made available on points of use for personnel to apply when necessary. • Reference values are also available on points of use to give guidance when performing activities in the analysis room. See attached document		
6.4.12	Practical measures are taken to prevent unintended adjustments of equipment from invalidating results.	• Instrument settings are locked (password,) and only authorized personnel may alter firmware/parameters. Any changes are recorded.		
6.4.13	Records are retained for equipment which can influence laboratory activities. The records shall include the following, where applicable:	• In the Equipment Master Log we record: unique ID,manufacturer's name model/serial,, calibration dates & results, modifications, and location. The register addresses points **B** through **H** • See attached equipment register		
	a) the identity of equipment, including software and firmware version;			
	b) the manufacturer's name, type identification, and serial number or other unique identification;			
	c) evidence of verification that equipment conforms with specified requirements;			
	d) the current location;			

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	e) calibration dates, results of calibrations, adjustments, acceptance criteria, and the due date of the next calibration or the calibration interval;			
	f) documentation of reference materials, results, acceptance criteria, relevant dates and the period of validity;			
	g) the maintenance plan and maintenance carried out to date, where relevant to the performance of the equipment;			
	h) details of any damage, malfunction, modification to, or repair of, the equipment.			
6.5	**Metrological traceability:** *How are the following addressed/implemented?*			
6.5.1 ***Note in Std.***	Establish and maintain metrological traceability of measurement results by means of a documented unbroken chain of calibrations, each contributing to the measurement uncertainty, linking them to an appropriate reference.	All measurement results in the laboratory are traceable to the International System of Units (SI) through an unbroken chain of calibrations, each with stated measurement uncertainties. Calibration certificates for balances, thermometers, probes, measuring equipment and furnaces are issued by SANAS-accredited laboratory, and the traceability chain is maintained in the laboratory's traceability memorandum. Certified reference weights traceable to SI units are used for balance verification with its calibrations conducted by a SANAS accredited facility, while standard reference materials (SRMs) and certified reference materials (CRMs) are employed for calorimetry and to ensure SI-based calibration. All traceability documentation, calibration certificates, and associated uncertainty budgets are reviewed, approved, and retained as objective evidence of compliance with ISO/IEC 17025 Clause 6.5. See attached Traceability memorandum. This Addresses **6.5.1** to **6.5.3**		
6.5.2 ***Notes in Std***	Measurement results are traceable to the International System of Units (SI) through one of the following:			
	a) calibration provided by a competent laboratory;			

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	a) certified values of certified reference materials provided by a competent producer with stated metrological traceability to the SI;			
	b) direct realization of the SI units ensured by comparison, directly or indirectly, with national or international standards.			
6.5.3	Demonstrate metrological traceability to an appropriate reference when metrological traceability to the SI units is not technically possible, e.g.			
	a) certified values of certified reference materials provided by a competent producer;			
	b) results of reference measurement procedures, specified methods or consensus standards that are clearly described and accepted as providing measurement results fit for their intended use and ensured by suitable comparison.			
7.2	**Selection, verification and validation of methods:** *How are the following addressed/implemented?*			
7.2.1	**Selection and verification of methods**			
7.2.1.1 ***Note in Std.***	Use appropriate methods and procedures for all laboratory activities and, where appropriate, for evaluation of the measurement uncertainty as well as statistical techniques for analysis of data.	• We adopt standard methods (SANS, ISO) where applicable (e.g., ISO 1928, SANS 5925). The CAB determines the UoM of each testing activity in its method validation reports, giving reference to TR26 requirements. • See attached method validation report as an example, AS002MV		

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7.2.1.2 ***Note in Std.***	All methods, procedures and supporting documentation, such as instructions, standards, manuals and reference data are kept up to date and made readily available to personnel	• We maintain a method control system; SOPs, manuals, software, and reference data are version controlled. Staff access the latest versions via the system folder at their point of use. Authorized personnel can only make changes to such documents. Method changes require review and notification before coming into effect. Refer to doc no 7.16		
7.2.1.3	When the customer does not specify the method to be used, the laboratory selects an appropriate method and inform the customer of the method chosen.	• If the client does not mandate a method, we propose a method (e.g., ISO 1171 for ash), inform the client in the test proposal, and document acceptance as reference in procedure 4.7.1 under section 3.4.		
	Use of methods published either in international, regional or national standards, or by reputable technical organizations, or in relevant scientific texts or journals, or as specified by the manufacturer recommended.	• The CAB uses Internationally recognized methods such ISO 562, 1928, 1171, 589, 13283, and SANS 5925 in its laboratory activities.		
	Laboratory-developed or modified methods can also be used.	• Quick ash, method validated and procedure available. • See procedure AS003IW & method validation AS003MV		
7.2.1.4	Methods verified before introducing them by ensuring that it can achieve the required performance. Records of the verification are retained. If the method is revised by the issuing body, verification is repeated to the extent necessary.	• All the SOP's are derived from recognized international standards. Moreover, the SOP's have been validated to ensure that each is fit for its intended use. • See procedure AS007IW & method validation AS002MV		

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7.2.1.5	When method development is required, it is a planned activity and is assigned to competent personnel equipped with adequate resources.	• Method development projects (e.g. for XRF analysis) are documented with objectives, timelines, assigned staff, resources, and periodic review. Changes require authorization and approval		
	Periodic review is carried out to confirm that the needs of the customer are still being fulfilled during method development. Any modifications to the development plan is approved and authorized.			
7.2.1.6 ***Note in Std.***	Deviations from methods are all documented, technically justified, authorized, and accepted by the customer.	• **How do we address such deviations within the system as this has to do with customers and yet we do not have any under ISO 17025?**		
7.2.2	**Validation of methods:** *How are the following addressed/implemented?*			
7.2.2.1 ***Note to Std.***	Validate non-standard methods, laboratory-developed methods and standard methods used outside their intended scope or otherwise modified. The validation to be as extensive as is necessary to meet the needs of the given application or field of application.	• All standardized methods employed, including in-house or modified methods (e.g. Quick Ash), we perform a full validation (precision, bias, linearity, robustness) and document the validation report, including statement of fitness for intended use • See method validation for Calorific Value, Volatile Matter for illustration. • The attached memorandum of conformity for this section outlines how the CAB has addresses validation of methods 7.2.21 through 7.2.2.4		

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7.2.2.2	Changes made to a validated method, the influence of such changes is determined and where they are found to affect the original validation, a new method validation to be performed.			
7.2.2.3 *Note to Std.*	Performance characteristics of validated methods as assessed for the intended use, is relevant to the customers' needs and consistent with specified requirements.			
7.2.2.4	Laboratory retains the following records of validation:			
	a) the validation procedure used;			
	b) specification of the requirements;			
	c) determination of the performance characteristics of the method;			
	d) results obtained;			
	e) statement on the validity of the method, detailing its fitness for the intended use.			
7.3	**Sampling:** *How are the following addressed/implemented?*			
7.3.1	Sampling plan and method available when lab/facility carries out sampling of substances, materials or products for subsequent testing or calibration.	• **As with our conformi**		
	Sampling method addresses the factors to be controlled to ensure the validity of subsequent testing or calibration results.			
	The sampling plan and method available at the site where sampling is undertaken. Sampling plans, whenever reasonable, be based on appropriate statistical methods.			
7.3.2	The sampling method describes:			

CLAUSE	ISO/IEC 17025:2017 REQUIREMENTS *How are the following addressed / implemented*	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
Note in Std.	a) the selection of samples or sites;			
	b) the sampling plan;			
	c) preparation and treatment of sample(s) from a substance, material or product to yield the required item for subsequent testing or calibration.			
7.3.3	Retains records of sampling data that forms part of the testing or calibration that is undertaken. These records include, where relevant:			
	a) reference to the sampling method used;			
	b) date and time of sampling;			
	c) data to identify and describe the sample (e.g. number, amount, name);			
	d) identification of the personnel performing sampling;			
	e) identification of the equipment used;			
	f) environmental or transport conditions;			
	g) diagrams or other equivalent means to identify the sampling location when appropriate;			
	h) deviations, additions to or exclusions from the sampling method and sampling plan.			
7.4	**Handling of test or calibration items:** *How are the following addressed/implemented?*			

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CLAUSE	ISO/IEC 17025:2017 REQUIREMENTS How are the following addressed / implemented	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
7.4.1	Procedure for the transportation, receipt, handling, protection, storage, retention, and disposal or return of test or calibration items.	• Material handling and storage procedure is available to give guidance for transportation, sample handling, retention and disposal of all materials that directly influence the quality and validity of results. • AS009IW is a waste management procedure that aids in managing laboratory waste and how to dispose it. **This addresses both the points below**		
	Precautions taken to avoid deterioration, contamination, loss or damage to the item during handling, transporting, storing/waiting, and preparation for, testing or calibration.			
	Handling instructions provided with the item are followed.			
7.4.2	Have a system for the unambiguous identification of test or calibration items.	• Each sample receives a unique ID (lab ID) upon receipt. All documentation and LIMS refer to the same ID.		

CLAUSE	ISO/IEC 17025:2017 REQUIREMENTS *How are the following addressed / implemented*	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
	Identification retained while the item is under the responsibility of the laboratory.	• Upon receiving a sample, it is given a unique ID number by LIMS software and retains the unique ID throughout the process until the sample is reported. The details of the sample can be retrieved long after its processing has been complete. There is segregation of parameters in the system which allows for easy retrieval of information pertaining any sample of interest • **Addresses the two points below**.		
	The system ensures that items will not be confused physically or when referred to in records or other documents.			
	The system, if appropriate, accommodate a sub-division of an item or groups of items and the transfer of items.			
7.4.3	Upon receipt of the test or calibration item, deviations from specified conditions is recorded.	• At sample receiving point, staff inspect sample condition. Any deviation (Bags and seal damage, weight deviations) is recorded, customer is consulted, and if they accept, a disclaimer is added in the final report. **Addresses the two points below.**		

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	When there is doubt about the suitability of an item for test or calibration, or when an item does not conform to the description provided, the laboratory consults the customer for further instructions before proceeding and record the results of this consultation.			
	When the customer requires the item to be tested or calibrated acknowledging a deviation from specified conditions, the laboratory include a disclaimer in the report indicating which results may be affected by the deviation.			
7.4.4	Items needed to be stored or conditioned under specified environmental conditions, these conditions are maintained, monitored and recorded.	• Samples requiring specific storage (e.g. CRM's and SRM's) placed in controlled environments, monitored (temp/humidity) and recorded. • See temperature recording form AS018TF		
7.5	**Technical records:** *How are the following addressed/implemented?*			
7.5.1	Technical records for each laboratory activity contain the results, report and sufficient information to facilitate, if possible, identification of factors affecting the measurement result and its associated measurement uncertainty and enable the repetition of the laboratory activity under conditions as close as possible to the original.	• We record and maintain every technical aspect of the laboratory that have direct influence on the quality and validity of results. These can be in a form of equipment verifications, calibration certificates, environmental monitoring and control (temperature & humidity), control charts, temperature recovery.		

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	The technical records include the date and the identity of personnel responsible for each laboratory activity and for checking data and results.	• The technical records include date, name of personnel, authorising personnel, technical number • Software (LIMS) has name of personnel conducting the activity, date & time. Each user has unique login credentials.		
	Original observations, data and calculations are recorded at the time they are made and are identifiable with the specific task.	• Software verifications are conducted bi-annually		
7.5.2	Amendments to technical records can be tracked to previous versions or to original observations.	• Technical records have version numbers, revision tables which explain the changes made to previous versions. For example, the version for this document was amended from AS001IW Rev 01 to AS001IW Rev 02		
	Both the original and amended data and files are kept, including the date of alteration, an indication of the altered aspects and the personnel responsible for the alterations.	• As per job descriptions, lab managers/supervisors onboarding the organization are tasked to develop, modify (where necessary) and validate methods employed in the laboratory, reviewing and authorization of results. Only personnel who satisfy competence and authorization criteria are allowed to perform or authorize work as outlined in each SOP. • See job descriptions: Lab Manager, Lab Supervisor		
7.6	**Evaluation of measurement uncertainty:** *How are the following addressed/implemented?*			

CLAUSE	ISO/IEC 17025:2017 REQUIREMENTS *How are the following addressed / implemented*	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
7.6.1	Identify the contributions to measurement uncertainty. When evaluating measurement uncertainty, all contributions which are of significance, including those arising from sampling, are taken into account using appropriate methods of analysis.	• Uncertainties associated with furnaces, balances, probes, calibrations are outlined in our uncertainty budget, see UB001		
7.6.2	Laboratory performing calibrations, including of its own equipment, shall evaluate the measurement uncertainty for all calibrations.			
7.6.3 ***Note in Std.***	The laboratory performing testing shall evaluate measurement uncertainty. Where the test method precludes rigorous evaluation of measurement uncertainty, an estimation shall be made based on an understanding of the theoretical principles or practical experience of the performance of the method.			
7.7	**Ensuring the validity of results:** *How are the following addressed/implemented?*			

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7.7.1	Have a procedure for monitoring the validity of results.	• The laboratory ensures the ongoing validity of results through a comprehensive quality control program that includes daily analysis of QC samples (reference coal), maintenance of control charts, evaluation of replicate analyses, and intermediate checks. Trends are reviewed when necessary to identify and flag any out-of-trend behaviour. The laboratory also participates quarterly in accredited coal proficiency testing, i.e., Coal Concept. All QC and PT outcomes are reviewed during management review meetings, and any breaches of established criteria—such as control chart exceedances—trigger formal investigations with documented corrective and preventive actions to ensure continuous reliability of results. • See procedure AS001V which addresses clause 7.7 in its entirety		
	The resulting data are recorded in such a way that trends are detectable and, where practicable, statistical techniques are applied to review the results.			
	Monitoring is planned and reviewed and include, where appropriate, but not be limited to:			
	a) use of reference materials or quality control materials;			
	b) use of alternative instrumentation that has been calibrated to provide traceable results;			

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	c) functional check(s) of measuring and testing equipment;			
	d) use of check or working standards with control charts, where applicable;			
	e) intermediate checks on measuring equipment;			
	f) replicate tests or calibrations using the same or different methods;			
	g) retesting or recalibration of retained items;			
	h) correlation of results for different characteristics of an item;			
	i) review of reported results;			
	j) intra-laboratory comparisons;			
	k) testing of blind sample(s).			
7.7.2 ***Note in Std.***	Monitor its performance by comparison with results of other laboratories, where available and appropriate.			
	Monitoring is planned and reviewed and include, but not be limited to, either or both of the following:			
	a) participation in proficiency testing;			

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	b) participation in inter-laboratory comparisons other than proficiency testing.			
7.7.3	Data from monitoring activities is analysed, used to control and, if applicable, improve the laboratory's activities. If the results of the analysis of data from monitoring activities are found to be outside pre-defined criteria, appropriate action is taken to prevent incorrect results from being reported.			
7.8	**Reporting of results:** *How are the following addressed/implemented?*			
7.8.1.1 ***Notes in Std.***	Results reviewed and authorized prior to release.	• All test reports undergo technical review by a technical signatory before issuing. Only authorized personnel may sign off reports • See attached procedure 7.8 Reporting of Results Procedure		
7.8.1.2	When agreed with the customer, the results may be reported in a simplified way. Any information listed in 7.8.2 to 7.8.7 that is not reported to the customer shall be readily available.	• There is a client receiving book in which clients outline which activities they need for their sample, which later aids in accurately reporting the correct parameters as requested.		
7.8.2	**Common requirements for reports (test, calibration or sampling)**			
7.8.2.1	Each report includes at least the following information, unless the laboratory has valid reasons for not doing so, thereby minimizing any possibility of misunderstanding or misuse:	• Procedure 7.8 Reporting of results addresses the requirements from a to p.		
	a) a title (e.g. "Test Report", "Calibration Certificate" or "Report of Sampling");			
	b) the name and address of the laboratory;			
	c) the location of performance of the laboratory activities, including when performed at a customer facility or at sites away from the laboratory's permanent facilities, or in associated temporary or mobile facilities;			

CLAUSE	ISO/IEC 17025:2017 REQUIREMENTS How are the following addressed / implemented	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
	d) unique identification that all its components are recognized as a portion of a complete report and a clear identification of the end;			
	e) the name and contact information of the customer;			
	f) identification of the method used;			
	g) a description, unambiguous identification, and, when necessary, the condition of the item;			
	h) the date of receipt of the test or calibration item(s), and the date of sampling, where this is critical to the validity and application of the results;			
	i) the date(s) of performance of the laboratory activity;			
	j) the date of issue of the report;			
	k) reference to the sampling plan and sampling method used by the laboratory or other bodies where these are relevant to the validity or application of the results;			
	l) a statement to the effect that the results relate only to the items tested, calibrated or sampled;			
	m) the results with, where appropriate, the units of measurement;			
	n) additions to, deviations, or exclusions from the method;			
	o) identification of the person(s) authorizing the report;			
	p) clear identification when results are from external providers.			

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CLAUSE	ISO/IEC 17025:2017 REQUIREMENTS *How are the following addressed / implemented*	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
7.8.2.2	Laboratory responsible for all the information provided in the report, except when information is provided by the customer. Data provided by a customer clearly identified. In addition, a disclaimer is put on the report when the information is supplied by the customer and can affect the validity of results. Where the laboratory has not been responsible for the sampling stage (e.g. the sample has been provided by the customer), it shall state in the report that the results apply to the sample as received.	• **This is also addressed in procedure 7.8 Reporting of results**		
7.8.3	**Specific requirements for test reports:** *How are the following addressed/implemented?*			
7.8.3.1	In addition to the requirements listed in 7.8.2, test reports shall, where necessary for the interpretation of the test results, include the following**:**			
	a) information on specific test conditions, such as environmental conditions;	• Addressed in the addendum section of the test report, under sampling plan. • See attached example of the test report, Pyrite		
	b) where relevant, a statement of conformity with requirements or specifications (see 7.8.6);	• Where relevant, we include environmental conditions, measurement uncertainty (if applicable or required), and disclaimers if deviations occurred • Opinions, Interpretations, statement of conformity based on decision rules are outside the scope of accreditation for this CAB, unless the client requests such; a disclaimer is issued for such situations to protect the CAB. This addresses C to E		

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CLAUSE	ISO/IEC 17025:2017 REQUIREMENTS How are the following addressed / implemented	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
	c) where applicable, the measurement uncertainty presented in the same unit as that of the measurand or in a term relative to the measurand (e.g. percent) when: — it is relevant to the validity or application of the test results; — a customer's instruction so requires, or — the measurement uncertainty affects conformity to a specification limit;			
	d) where appropriate, opinions and interpretations (see 7.8.7);			
	e) additional information which may be required by specific methods, authorities, customers or groups of customers			
7.8.3.2	Laboratory responsible for the sampling activity, its test reports shall meet the requirements listed in 7.8.5 where necessary for the interpretation of test results.	• **`This is addressed in procedure AS008IW**		
7.8.4	**Specific requirements for calibration certificates:** *How are the following addressed/implemented?*			
7.8.4.1 ***Note in Std.***	In addition to the requirements listed in 7.8.2, calibration certificates shall include the following:	**This section seems to be directed to calibration laboratories. Since the CAB is a testing lab, it does not produce any calibration certificates. However, we have checked the requirements as outlined from a-f, 7.8.4.2 and 7.8.4.3. The external service provider for calibration of the CAB's equipment Does conform to clause 7.8.4 in its entirety.**		
	a) the measurement uncertainty of the measurement result presented in the same unit as that of the measurand or in a term relative to the measurand (e.g. percent);			
	b) the conditions (e.g. environmental) under which the calibrations were made that have an influence on the measurement results;			

CLAUSE	ISO/IEC 17025:2017 REQUIREMENTS *How are the following addressed / implemented*	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
	c) a statement identifying how the measurements are metrologically traceable (see Annex A);			
	d) the results before and after any adjustment or repair, if available;			
	e) where relevant, a statement of conformity with requirements or specifications (see 7.8.6);			
	f) where appropriate, opinions and interpretations (see 7.8.7).			
7.8.4.2	Laboratory responsible for the sampling activity, its calibration certificates shall meet the requirements listed in 7.8.5 where necessary for the interpretation of test results.			
7.8.4.3	A calibration certificate or calibration label shall not contain any recommendation on the calibration interval except where this has been agreed with the customer.			
7.8.5	**Reporting sampling – specific requirements:** *How are the following addressed/implemented?*			
	Where the laboratory is responsible for the sampling activity, in addition to the requirements listed in 7.8.2, reports shall include the following, where necessary for the interpretation of results:	• **As stated in the statement of conformity, the CAB only does sampling when requested, however, sampling will not form part of the CAB's scope of accreditation, see attached statement of conformity & sampling procedure**		
	a) the date of sampling;			
	b) unique identification of the item or material sampled (including the name of the manufacturer, the model or type of designation and serial numbers as appropriate);			
	c) the location of sampling, including any diagrams, sketches or photographs;			

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CLAUSE	ISO/IEC 17025:2017 REQUIREMENTS *How are the following addressed / implemented*	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
	d) a reference to the sampling plan and sampling method;			
	e) details of any environmental conditions during sampling that affect the interpretation of the test results;			
	f) information required to evaluate measurement uncertainty for subsequent testing or calibration.			
7.8.6	**Reporting statements of conformity:** *How are the following addressed/implemented?*			
7.8.6.1 ***Note in Std***	When a statement of conformity to a specification or standard is provided, the laboratory documents the decision rule employed, taking into account the level of risk (such as false accept and false reject and statistical assumptions) associated with the decision rule employed and apply the decision rule.	• The laboratory normally reports measured results only. Statements of conformity (e.g. "Pass/Fail", "Meets specification") are not routinely provided unless explicitly requested by the client or required by a standard or contract. When such a statement is required, the laboratory will apply a documented decision rule that takes into account measurement uncertainty and the associated risk of false acceptance or false rejection, as described in SOP AS010IW "Conformity Statements & Decision Rules." • Procedure attached Addresses 7.8.6.2 as well.		
7.8.6.2 ***Note in Std.***	Report on the statement of conformity clearly identifies:			
	a) to which results the statement of conformity applies;			
	b) which specifications, standards or parts thereof are met or not met;			

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CLAUSE	ISO/IEC 17025:2017 REQUIREMENTS How are the following addressed / implemented	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
	c) the decision rule applied (unless it is inherent in the requested specification or standard).			
7.8.7	**Reporting opinions and interpretations**: *How are the following addressed/implemented?*			
7.8.7.1 ***Note in Std.***	When opinions and interpretations are expressed, the laboratory to ensure that only personnel authorized for the expression of opinions and interpretations releases the respective statement. Laboratory document the basis upon which the opinions and interpretations have been made.	• The CAB's scope of accreditation does not cover opinions and interpretations. However, when requested by clients, a disclaimer has to be given to the client. • This coveres 7.8.7 in its entirety		
7.8.7.2	Opinions and interpretations expressed in reports are based on the results obtained from the tested or calibrated item and are clearly identified as such.			
7.8.7.3	Opinions and interpretations directly communicated by dialogue with the customer, a record of the dialogue is retained.			
7.8.8	**Amendments to reports:** *How are the following addressed/implemented?*			
7.8.8.1	When an issued report needs to be changed, amended or re-issued, any change of information is clearly identified and, where appropriate, the reason for the change included in the report.	• Amended reports must be clearly marked, state the reason for change, retain revision control, and be re-authorized, see procedure ASMF 7.8 section 4.7		
7.8.8.2	Amendments to a report after issue are made only in the form of a further document, or data transfer, which includes the statement "Amendment to Report, serial number... [or as otherwise identified]", or an equivalent form of wording.	• Amended report will have a title change from "Test Report" to "Amended Test Report"		

CLAUSE	ISO/IEC 17025:2017 REQUIREMENTS How are the following addressed / implemented	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
7.8.8.3	A completely new report is uniquely identified and contains a reference to the original that it replaces.	• The new report will keep the original report number but with an "RI" at the end to show that it is a re-issue (e.g., Azuresky no 135 RI)		

Additional / General Comments *This space may also be used to expand on comments in specific sections*

Team Leader's Signature:		**Date:**	

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